

The Optimal Bourgain Constant in Quantitative Bochner Weak Precompactness

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Source and open question

The source paper is Jose Rodriguez, “ ε -weakly precompact sets in Banach spaces”, arXiv:2102.01546, submitted 2021-02-02. The open problem addressed here is Question 2.21.

Proof of Theorem 2.11. (ii) follows from (i), Theorem 2.1(i) and Lemma 2.5.

We begin the proof of (i) with the following:

Particular case. Suppose that $f_n \in L_\infty(\mu, X)$ for all $n \in \mathbb{N}$ and that

$$\sup_{n \in \mathbb{N}} \|f_n\|_{L_\infty(\mu, X)} < \infty.$$

Write $Y := L_1([0, 1], X)$. For each $n \in \mathbb{N}$ we define (thanks to Lemma 2.19) $h_n \in L_\infty(\mu, Y)$ by

$$h_n(\omega) := r_n \otimes f_n(\omega) \quad \text{for } \mu\text{-a.e. } \omega \in \Omega.$$

By the assumption on F and Lemma 2.17, we have

$$\text{clust}_{Y^{**}}(h_n(\omega)) \subseteq \varepsilon B_{Y^{**}} \quad \text{for } \mu\text{-a.e. } \omega \in \Omega.$$

Bearing in mind that $\sup_{n \in \mathbb{N}} \|h_n\|_{L_\infty(\mu, Y)} < \infty$, an appeal to Lemma 2.18 ensures that

$$\text{clust}_{L_1(\mu, Y)^{**}}(h_n) \subseteq \varepsilon B_{L_1(\mu, Y)^{**}}.$$

Write $Z := L_1(\mu, X)$ and let $\Phi : L_1(\mu, Y) \rightarrow L_1([0, 1], Z)$ be the natural isometric isomorphism. Then $\Phi(h_n) = r_n \otimes f_n$ for all $n \in \mathbb{N}$ and therefore

$$\text{clust}_{L_1([0, 1], Z)^{**}}(r_n \otimes f_n) \subseteq \varepsilon B_{L_1([0, 1], Z)^{**}}.$$

Therefore, $(f_n)_{n \in \mathbb{N}}$ cannot be an ℓ_1 -sequence with constant $C > \varepsilon$ (by Lemma 2.16), as required.

We now turn to the:

General case. Fix $C > \varepsilon$ and choose any $0 < \eta < C - \varepsilon$. Since $(f_n)_{n \in \mathbb{N}}$ is uniformly integrable, Lemma 2.20 gives a sequence $(A_n)_{n \in \mathbb{N}}$ in Σ such that each $g_n := f_n \chi_{A_n}$ belongs to $L_\infty(\mu, X)$ and:

- $\sup_{n \in \mathbb{N}} \|g_n\|_{L_\infty(\mu, X)} < \infty$,
- $\|f_n - g_n\|_{L_1(\mu, X)} \leq \eta$ for every $n \in \mathbb{N}$.

Define a multi-function $\tilde{F} : \Omega \rightarrow \mathcal{P}(X)$ by $\tilde{F}(\omega) := F(\omega) \cup \{0\}$ for every $\omega \in \Omega$. By the *Particular case*, the sequence $(g_n)_{n \in \mathbb{N}}$ in $S_1(\tilde{F})$ cannot be an ℓ_1 -sequence with constant $C - \eta$. Since $\|f_n - g_n\|_{L_1(\mu, X)} \leq \eta$ for all $n \in \mathbb{N}$, we conclude that $(f_n)_{n \in \mathbb{N}}$ cannot be an ℓ_1 -sequence with constant C . The proof is finished. $\square$

Question 2.21. *Is constant 2 optimal in Theorem 2.11?*

The concept of $\delta\mathcal{C}$ -sets in Lebesgue-Bochner spaces (which corresponds to the case $\varepsilon = 0$ of the following definition) appeared first in [8] and was discussed further in [6].

The question asks whether the constant 2 in Rodriguez's Theorem 2.11 is optimal. In the notation of that theorem, (Ω, Σ, μ) is a probability space, X is a Banach space, $F : \Omega \rightarrow \mathcal{P}(X)$ has

$F(\omega)$ ε -weakly precompact for almost every ω , and

$$S_1(F) = \{f \in L_1(\mu, X) : f(\omega) \in F(\omega) \text{ for a.e. } \omega\}.$$

The source theorem proves that every uniformly integrable subset of $S_1(F)$ is 2ε -weakly precompact.

Result

Theorem 1. *In Theorem 2.11 of the source paper the constant 2 can be replaced by 1. More precisely, if $F(\omega)$ is ε -weakly precompact for almost every ω , then:*

- (i) *no uniformly integrable sequence in $S_1(F)$ is an ℓ_1 -sequence with constant $C > \varepsilon/2$;*
- (ii) *every uniformly integrable subset of $S_1(F)$ is ε -weakly precompact in $L_1(\mu, X)$.*

Consequently, for every ε -weakly precompact set $M \subseteq X$, the set $L(M)$ is ε -weakly precompact in $L_1(\mu, X)$.

Theorem 2. *The constant 1 in Theorem 1 is optimal. In particular, Question 2.21 has a negative answer: the constant 2 is not optimal, and the best universal constant is 1.*

Definitions used

For a bounded sequence (x_n) in a Banach space X , let

$$\delta(x_n) = \text{diam clust}_{X^{**}}(x_n) = \sup_{x^* \in B_{X^*}} \left(\limsup_n x^*(x_n) - \liminf_n x^*(x_n) \right).$$

A bounded set $M \subseteq X$ is ε -weakly precompact if every sequence in M has a subsequence (x_{n_k}) with $\delta(x_{n_k}) \leq \varepsilon$. An ℓ_1 -sequence with constant C is a bounded sequence (x_n) satisfying

$$\left\| \sum_{j=1}^m a_j x_j \right\| \geq C \sum_{j=1}^m |a_j|$$

for all finite scalar families.

The midpoint lemma

The source proof of Lemma 2.17 uses Rodriguez's Lemma 2.14 with the centre $\liminf h_n$. The factor 2 is lost exactly there. The following variant centres at the midpoint of the scalar cluster interval.

Lemma 1. *Let (Ω, Σ, μ) be a probability space, let (h_n) be a bounded sequence in $L_\infty(\mu)$, and suppose*

$$\limsup_n h_n - \liminf_n h_n \leq \varepsilon \quad \text{a.e.}$$

Put

$$m = \frac{1}{2} \left(\limsup_n h_n + \liminf_n h_n \right) \in L_\infty(\mu).$$

Then

$$\limsup_n \int_\Omega |h_n - m| d\mu \leq \frac{\varepsilon}{2}.$$

Proof. Let $g_n = |h_n - m|$. The sequence (g_n) is bounded in $L_\infty(\mu)$. Pointwise,

$$\limsup_n g_n \leq \frac{1}{2} \left(\limsup_n h_n - \liminf_n h_n \right) \leq \frac{\varepsilon}{2}.$$

Rodriguez's Lemma 2.13, the reverse Fatou inequality for bounded L_∞ -sequences on a probability space, gives

$$\limsup_n \int_\Omega g_n d\mu \leq \int_\Omega \limsup_n g_n d\mu \leq \frac{\varepsilon}{2}.$$

□

Improved Rademacher lemma

Let (r_n) be the Rademacher functions on $[0, 1]$.

Lemma 2. *Let Z be a Banach space, let $M \subseteq Z$ be ε -weakly precompact, and let (z_n) be a sequence in M . Then*

$$\text{clust}_{L_1([0,1],Z)^{**}}(r_n \otimes z_n) \subseteq \frac{\varepsilon}{2} B_{L_1([0,1],Z)^{**}}.$$

Proof. The proof is Rodriguez's Lemma 2.17 with Lemma 1 substituted for Lemma 2.14. Let F be a weak-star cluster point of $(r_n \otimes z_n)$ and fix $\varphi \in B_{L_1([0,1],Z)^*}$. Represent φ by a weak-star scalarly measurable function $[0, 1] \rightarrow Z^*$ with $\|\varphi(t)\| \leq 1$ a.e. Choose a subsequence (n_k) such that

$$\langle r_{n_k} \otimes z_{n_k}, \varphi \rangle \rightarrow \langle F, \varphi \rangle.$$

Passing to a further subsequence, using ε -weak precompactness of M , we may suppose $\delta(z_{n_k}) \leq \varepsilon$.

Set

$$a_k(t) = \langle z_{n_k}, \varphi(t) \rangle.$$

Then (a_k) is bounded in $L_\infty[0, 1]$, and for almost every t ,

$$\limsup_k a_k(t) - \liminf_k a_k(t) \leq \delta(z_{n_k}) \leq \varepsilon.$$

Let

$$m(t) = \frac{1}{2} \left(\limsup_k a_k(t) + \liminf_k a_k(t) \right).$$

By Lemma 1,

$$\limsup_k \int_0^1 |a_k(t) - m(t)| dt \leq \frac{\varepsilon}{2}.$$

Since (r_{n_k}) is weakly null in $L_1[0, 1]$ and $m \in L_\infty[0, 1]$,

$$\int_0^1 r_{n_k}(t) m(t) dt \rightarrow 0.$$

Thus

$$|\langle r_{n_k} \otimes z_{n_k}, \varphi \rangle| \leq \left| \int_0^1 r_{n_k}(t) m(t) dt \right| + \int_0^1 |a_k(t) - m(t)| dt,$$

and taking $\limsup_k$ yields $|\langle F, \varphi \rangle| \leq \varepsilon/2$. Since φ was arbitrary in the dual unit ball, $\|F\| \leq \varepsilon/2$. □

Proof of Theorem 1

First assume the particular bounded case used in Rodriguez's proof: each $f_n \in L_\infty(\mu, X)$ and $\sup_n \|f_n\|_{L_\infty(\mu, X)} < \infty$. Let $Y = L_1([0, 1], X)$ and define

$$h_n(\omega) = r_n \otimes f_n(\omega) \in Y.$$

For almost every ω , Lemma 2 applied to the sequence $(f_n(\omega))$ in $F(\omega)$ gives

$$\text{clust}_{Y^{**}}(h_n(\omega)) \subseteq \frac{\varepsilon}{2} B_{Y^{**}}.$$

Rodriguez's Lemma 2.18 then gives

$$\text{clust}_{L_1(\mu, Y)^{**}}(h_n) \subseteq \frac{\varepsilon}{2} B_{L_1(\mu, Y)^{**}}.$$

Under the natural isometric isomorphism $L_1(\mu, L_1([0, 1], X)) \cong L_1([0, 1], L_1(\mu, X))$, this says

$$\text{clust}_{L_1([0, 1], L_1(\mu, X))^{**}}(r_n \otimes f_n) \subseteq \frac{\varepsilon}{2} B.$$

Rodriguez's Lemma 2.16 states that if (f_n) is an ℓ_1 -sequence with constant C , then $(r_n \otimes f_n)$ is an ℓ_1 -sequence with the same constant C , and in particular its bidual cluster set is not contained in ηB for any $\eta < C$. Therefore the bounded particular case cannot contain an ℓ_1 -sequence with constant $C > \varepsilon/2$.

Now let (f_n) be any uniformly integrable sequence in $S_1(F)$ and suppose, for contradiction, that it is an ℓ_1 -sequence with constant $C > \varepsilon/2$. Choose $0 < \eta < C - \varepsilon/2$. Rodriguez's truncation Lemma 2.19 gives measurable sets A_n such that $g_n = f_n \chi_{A_n}$ is uniformly L_∞ -bounded and

$$\|f_n - g_n\|_{L_1(\mu, X)} \leq \eta \quad (n \in \mathbb{N}).$$

The sequence (g_n) belongs to $S_1(\tilde{F})$ for $\tilde{F}(\omega) = F(\omega) \cup \{0\}$, whose values are still ε -weakly precompact. Moreover (g_n) is an ℓ_1 -sequence with constant at least $C - \eta > \varepsilon/2$. This contradicts the bounded case. Hence part (i) follows.

For part (ii), let $W \subseteq S_1(F)$ be uniformly integrable. Fix $\varepsilon' > \varepsilon$. If W were not ε' -weakly precompact, Theorem 2.1(i) of the source paper would give an ℓ_1 -sequence in W with constant $\varepsilon'/2 > \varepsilon/2$, contradicting part (i). Thus W is ε' -weakly precompact for every $\varepsilon' > \varepsilon$, and Rodriguez's Lemma 2.5 implies that W is ε -weakly precompact.

Proof of optimality

Let $X = \ell_1$ and let $M = \{\alpha e_n : n \in \mathbb{N}\}$ for some $\alpha > 0$. In the one-point probability space, $L_1(\mu, X)$ identifies with X and $L(M) = M$.

The set M is 2α -weakly precompact. Indeed, every sequence in M has either a constant subsequence, with $\delta = 0$, or a subsequence of distinct basis vectors. For a distinct basis subsequence, the diameter is exactly 2α : it is at most 2α by norm diameter, and it is at least 2α by evaluating against a bounded scalar sequence in ℓ_∞ which alternates between 1 and -1 along the chosen indices.

Thus M is not η -weakly precompact for any $\eta < 2\alpha$. If a universal constant $K < 1$ replaced the constant 1 in Theorem 1, applying it with $\varepsilon = 2\alpha$ would force $L(M) = M$ to be $2K\alpha$ -weakly precompact, contradicting $2K\alpha < 2\alpha$. Hence no constant smaller than 1 is possible. The same example shows the threshold $C > \varepsilon/2$ in part (i) is sharp.

Novelty and literature check

Before promotion, bounded searches were run on 2026-06-13 for exact and close phrases involving:

- Question 2.21, 2102.01546, and `epsilon-weakly precompact`;
- Theorem 2.11, constant 2, and Bourgain;
- Bourgain–Maurey–Pisier, weakly conditionally compact sets in $L_1 X$, and Bourgain’s averaging result for ℓ_1 -sequences.

The searches found Rodriguez’s arXiv paper and background around Bourgain, Maurey–Pisier, Behrends, Gasparis, and Kalenda–Pfitzner–Spurny, but no separate later source explicitly resolving Question 2.21 or advertising the constant-one improvement. This is a bounded novelty check, not a claim of exhaustive bibliographic certainty.

Verification notes and human-review focus

The proof is short and the review focus should be correspondingly narrow:

1. Check Lemma 1: it uses only the source paper’s reverse Fatou Lemma 2.13 for bounded L_∞ sequences.
2. Check that the midpoint function m in Lemma 2 is measurable and in L_∞ ; this follows because it is built from $\limsup$ and $\liminf$ of bounded measurable scalar functions.
3. Check that the passage from the pointwise improved Rademacher lemma to the global Bochner statement is exactly the source proof of Theorem 2.11, with ε replaced by $\varepsilon/2$ up to the final Behrends endpoint argument.
4. Check the one-point ℓ_1 example for optimality.

Current verdict: likely valid candidate full solution, pending human review.